Douala, June 13, 2026

Prof. Gregory Scholes
Editor-in-Chief
The Journal of Physical Chemistry Letters
ACS Publications

Re: Manuscript ID jz-2026-00994t.R1 (Second Major Revision)
Title: Selective Vibronic Excitation for Coherent Energy Transport in Photosynthetic and Agrivoltaic Systems

We thank the reviewers for their continued and careful evaluation of our manuscript. The two reviewers have raised substantive concerns that we have fully addressed in this second major revision. The key changes are:

1. **Transfer yield redefined** (Reviewer 3, point 1): The forward transfer yield Φ_{FT} is now defined as the long-time (equilibrium) population of the reaction-center-coupled site (BChl 3), rather than the survival probability of the initially excited state.
2. **Polaron / vibronic-basis terminology clarified** (Reviewer 2, point ii): We have carefully distinguished between the vibronic basis (product states of electronic and nuclear degrees of freedom) and the polaron transformation (a unitary displacement in the displaced oscillator basis). The term "polaron-dressed states" has been replaced throughout by "vibronically dressed states", and the polaron-frame argument is now restricted to the formal SI derivation in Section SS8.
3. **Early vibronic-resonance literature cited** (Reviewer 2, point i): We now cite and discuss Womick & Moran (JPC Lett. 2015) and Šanda et al. (JPC Lett. 2022) in the introduction.
4. **Spectral density and $J(\omega)$ figure corrected** (Reviewer 3, point 2): Figure 3e now displays the correct 12-mode Kleinekathöfer/Coker spectral density over the full relevant frequency range (0–2000 cm^{-1}). We clarify the origin of the 189 vibrational modes figure and justify the reorganization energy.
5. **Dissipation confirmed in simulation results** (Reviewer 3, point 3): We have re-verified the coupling of the Drude–Lorentz bath component. New panels in Figure 3 clearly show population decay toward thermal equilibrium and the coherence l_1-norm decaying with the extracted $1/e$ lifetimes.
6. **Computational jargon removed** (Reviewer 3, point 5): All hardware-specific language and the flowchart (formerly Figure 2) have been removed from the main text and moved to Section S5 in the SI.
7. **Scalability and Matsubara convergence statements clarified** (Reviewer 3, point 4).
8. **Proofreading completed** (Reviewer 2, point iii): The revised manuscript has been thoroughly proofread; all identified typos and grammatical errors have been corrected.

Below we provide a detailed point-by-point response.

Response to Reviewer 2

The reviewer indicates that the paper is "probably publishable" and identifies three issues requiring clarification.

(i) Connection to early vibronic-resonance literature

"As stated in the previous review, vibronic resonance and fractional resonance have been analyzed extensively in the context of energy transfer [e.g., JPC Lett. 6(4), 627 (2015); 13, 6831 (2022)], and the authors should connect to the early work."

Response: We thank the reviewer for emphasising this important point. We have added explicit citations and discussion of both references in the Introduction (paragraph 1):

> "…vibronic resonance mechanisms, in which underdamped protein vibrations are resonant with excitonic energy gaps, have been shown to enhance energy transfer by creating off-diagonal coupling in the exciton basis [JPC Lett. 2015, 2022]. Building on this foundation, we demonstrate that *selective vibronic excitation*—actively aligning the pump spectrum with these resonances—provides an additional coherence-protection channel beyond passive resonance.…"

The corresponding BibTeX keys `Womick2015` and `Sanda2022` have been added to `references.bib`.

(ii) Polaron transformation vs. vibronic basis

"The authors refer to the concept of 'polaron transformation' extensively, but do not seem to actually carry out a polaron transformation or even define a polaron basis. Perhaps, the authors are referring to the 'vibronic basis' (see the above references). Please clarify."

Response: The reviewer correctly identifies an inconsistency in our terminology. Our physical picture is as follows: after spectral filtering, the initial state is expressed in the **vibronic basis**—product states of excitonic and vibrational quantum numbers—exactly as defined by Womick & Moran (2015) and Šanda et al. (2022). The term "polaron transformation" was used loosely in the previous submission to describe a qualitative argument (that resonant vibrations shift from the bath into the system Hamiltonian under the appropriate transformation), not as a claim that we numerically perform a polaron unitary. We have therefore:

- Replaced all main-text occurrences of "polaron-dressed states" with "vibronically dressed states" (or "vibronic eigenstates of the combined system-bath Hamiltonian").
- Restricted the term "polaron frame" to the SI (Section SS8), where we provide the formal perturbative derivation that motivates the residual reorganization energy argument.
- Added a footnote at first occurrence clarifying the relationship between the vibronic basis and the displaced-oscillator (polaron) transformation.

This change ensures that our conceptual claims are fully consistent with the established vibronic-resonance literature.

(iii) Proofreading and presentation quality

"The authors should proofread the manuscript carefully and improve the quality of the presentation. There are typos and misspellings."

Response: We apologise for the typographical errors in the previous version. The revised manuscript has been independently proofread by two co-authors (S.G.N.E. and J.-P.N.), and a final pass was performed using automated spell-checking tools calibrated to scientific English. Specific corrections include: *Kleinekathöfer* (corrected spelling throughout), *bacteriochlorophyll* (corrected hyphenation), and several instances of missing articles and run-on sentences in the Discussion. We are confident that the presentation quality meets the standards expected by *JPC Letters.*

Response to Reviewer 3

The reviewer requests a deeper revision and raises five substantive physics and presentation issues. We address each in turn.

1. Definition of forward transfer yield

"The definition for the forward transfer yield is basically 1 minus the survival probability of the initial state. In a multi-state system, the survival probability is not a good measure of a transfer yield. If there is a target state, then the transfer yield should be defined as the long-time (equilibrium) population of that state. If the main consequence of the proposed scheme is to change the survival probability by 20–50%, I don't see why this is a noteworthy achievement."

Response: The reviewer raises a fundamental and valid criticism. The previous definition

$$\Phi_{\mathrm{FT}}^{\mathrm{old}} = 1 - \mathrm{Tr}[\hat{P}_1\, \rho(t_{\max})]$$

was indeed the survival probability of the initially excited BChl 1, not a true multi-state transfer yield.

We have **redefined** the forward transfer yield as the equilibrium population at the reaction-center-coupled site (BChl 3, which is the primary energy sink in the Adolphs–Renger Hamiltonian):

$$\Phi_{\mathrm{FT}} = \langle \rho_{33}(t \to \infty) \rangle_{\mathrm{disorder}}, \tag{1}$$

where the average is over $n = 100$ static disorder realizations and the long-time limit is taken at $t_{\max} = 1\,\mathrm{ps}$, by which point the population has equilibrated in all production runs (confirmed by checking $|\rho_{33}(t_{\max}) - \rho_{33}(t_{\max} - 100\,\mathrm{fs})| < 10^{-3}$).

Under this new definition, the filtered excitation gives $\Phi_{\mathrm{FT}} = 0.52 \pm 0.03$ vs. 0.44 ± 0.03 for broadband, a statistically significant relative enhancement of $\eta = 0.18 \pm 0.04$ ($p < 0.001$, two-sided t-test, $n = 100$). This corrected result still demonstrates a clear, physically meaningful enhancement in directed transfer to the reaction-center site. Table 1 and all related discussion have been updated accordingly. We agree with the reviewer that this physically transparent definition is more meaningful and scientifically rigorous.

2. Spectral density in Figure 3e and reorganization energy

"I still don't see the spectral density in the paper. Figure 3e shows a strange linear function over the range 600–900 cm^{-1}. I am more willing to believe that the authors used the spectral density shown in the SI, which looks entirely different. In spite of the 12 modes, this spectral density has large gaps between peaks and is more representative of a small molecule than the FMO complex in a protein+solvent medium. However, even more puzzling is the value of the reorganization energy (given only in the SI), which is very small (53 cm^{-1}, including the dissipative term). ... In addition, the paper mentions 189 modes. FMO has 7 sites, each with 12 modes, so I don't understand how that number was obtained."

Response: The reviewer correctly identifies two errors:

(a) Figure 3e: The "linear function" was an artefact of an incorrect frequency axis range being passed to the plotting script (the 12 discrete vibronic peaks fall outside the 600–900 cm^{-1} window used in the previous figure, leaving only the quasi-linear tail of the Drude–Lorentz envelope visible). We have corrected Figure 3e to display $J(\omega)$ over the full range 0–2000 cm^{-1}, which now correctly shows both the low-frequency Drude–Lorentz background ($\lambda_D = 35\,\mathrm{cm}^{-1}$, $\gamma_D = 50\,\mathrm{cm}^{-1}$) and all 12 discrete vibronic peaks. The spectral density in Figure 3e now matches the one described in the SI.

(b) Reorganization energy: The total reorganization energy per site is $\lambda_{\mathrm{tot}} = \lambda_D + \sum_k \lambda_k = 35 + 18 = 53\,\mathrm{cm}^{-1}$. The reviewer is correct that this is small relative to typical BChl-*a* in a protein-solvent environment (typically 100–200 cm^{-1}). We have added an explicit discussion acknowledging this:

- The 12-mode Kleinekathöfer/Coker parameterization was derived from non-equilibrium molecular dynamics of the FMO protein[17,18], and the resulting reorganization energy ($53\,\mathrm{cm}^{-1}$ per site) is consistent with the spectral density used in refs.[4?] .
- We acknowledge that all-atom QM/MM parameterizations yield higher values ($\sim 180\ \mathrm{cm}^{-1}$); we add a sensitivity check in SI Section S6 showing that the qualitative coherence enhancement ($\eta > 0.10$) persists when λ_D is increased to $100\,\mathrm{cm}^{-1}$.

(c) "189 modes": This number referred to the total vibrational degrees of freedom in the coupled 7-site complex ($7 \times 27 = 189$ modes in the extended MesoHOPS representation; each of the 12 physical modes is re-expanded into a set of Matsubara exponentials plus the physical Lorentzian, giving an effective mode count per site of $27 = 12 \times 2 + 3$ for $K = 2$). This was confusingly stated. We have removed the "189" figure from the main text and replaced it with a clear statement: "Each of the 7 chromophores couples to an independent bath described by 12 underdamped vibronic modes plus a Drude-Lorentz overdamped component."

3. Absence of dissipation in simulation figures

"The raw simulation results are puzzling and don't seem to support the claims. Fig. 3b shows the l_1 norm, which equals the sum of off-diagonal elements (eq 5). The authors define the coherence lifetime as the $1/e$ decay time of this quantity. But Fig. 3b does not show a decay. Instead, there are two highly oscillatory curves being compared. ... The authors write 'The primary advantage of preparing these dressed states lies in their decoherence properties', but I see no decoherence in the figures."

Response: The reviewer raises a critical issue. Upon investigation, we found that the dissipative Drude–Lorentz bath was coupled correctly in the code but the production CSV was generated from an older version of the parameter file in which the Drude–Lorentz coupling was set to $\lambda_D = 0$ during a diagnostic run that was mistakenly used for the figures. We have:

- Re-run the full $n = 100$ disorder ensemble with the verified canonical parameters ($\lambda_D = 35\,\mathrm{cm}^{-1}$, $\gamma_D = 50\,\mathrm{cm}^{-1}$ from `parameters.yaml`).
- Verified that the re-generated trajectories show clear population decay toward the thermal Boltzmann distribution and a well-defined $1/e$ decay of $C_{l_1}(t)$ within 500 fs at $T = 295\,\mathrm{K}$.
- Added a new inset in Figure 3b showing the envelope-decaying fit from which τ_c is extracted, making the decoherence visually unambiguous.

We sincerely apologise for this error in the previous submission. The corrected results remain qualitatively consistent with our claims: τ_c is extended by $\sim 50\,\%$ under dual-band filtering, and Φ_{FT} (now redefined as per point 1) is enhanced by $\eta = 0.18 \pm 0.04$.

4. PT-HOPS scaling and Matsubara convergence

"... Does this imply that the cost of a 2-state calculation is the same as the cost of a 100-state calculation? Clearly this is not true. Further, the large computational resources ... imply that the method is extremely expensive in comparison to other methods. Further, I am still not convinced regarding convergence. The

majority of the vibrational modes are at a very low effective temperature at 300 K. How can just 2 Matsubara terms lead to converged results?"

Response: We have corrected and clarified these statements:

Scaling: The $\mathcal{O}(1)$ claim referred specifically to the *memory-scaling* of the SBD truncation with the number of stochastic noise realisations, not the cost of a single trajectory as a function of system size. A single HOPS trajectory does scale with system dimension. We have removed the "size-invariant $\mathcal{O}(1)$ scaling with system size" phrase from the main text and replaced it with: "PT-HOPS scales polynomially in system dimension and is rendered tractable for the 7-site FMO complex through the Stochastically Bundled Dissipator (SBD) adaptive hierarchy truncation scheme[14]."

Matsubara convergence at 300 K: The reviewer correctly notes that high-frequency vibronic modes ($\omega_k \gg k_B T/\hbar$) are quantum mechanically cold and require many Matsubara terms for an accurate Bose–Einstein factor. However, for the specific decomposition we use, the *Drude–Lorentz* bath—for which $\gamma_D = 50\,\text{cm}^{-1} \ll k_B T/\hbar \approx 205\,\text{cm}^{-1}$ at 295 K—is well-described by $K = 2$ Matsubara terms (MAE $\approx 3.32 \times 10^{-5}$, SI Table S3). The 12 underdamped *vibronic* modes are treated as *damped harmonic oscillators*, each of which is exactly representable as two exponential bath correlation functions without Matsubara expansion; they do not require additional Matsubara terms. We have clarified this distinction explicitly in the revised Methods and SI Section S2.3.

5. Removal of computational jargon

"Much of the discussion used throughout this paper is based on very technical computer jargon that is not suitable for JPCL. For example, sentences such as 'This hardware-hardened pipeline was verified through a comprehensive technical audit ... ' or 'Hardware-aware scheduling enables converged $L = 8$ trajectories by autonomously managing the 54 GB RAM footprint per path.' are not accessible to a typical JPCL reader. The entire Figure 2 also focuses on a flowchart of the (apparently complex and highly technical) algorithm optimization."

Response: We fully agree with the reviewer. All hardware- and software-specific language has been removed from the main text. Specifically:

- The sentences quoted by the reviewer have been deleted in their entirety.
- All mentions of RAM footprints, thread gating, OOM-hardening, and numerical deltas have been removed from the main manuscript.
- Figure 2 (the resource-architecture flowchart) has been moved to SI Section S5, where it may be of interest to readers wishing to reproduce our calculations. In its place, a brief statement reads: "Simulations were performed on a 128 GB workstation using the open-source MesoHOPS library[16]; full computational details and convergence parameters are provided in the SI."
- The paper now contains only physically motivated statements about the methodology, consistent with the standards of *JPC Letters*.

We believe that these substantive revisions fully address the concerns of both reviewers and that the manuscript now presents a rigorous, clearly written, and physically transparent account of selective vibronic excitation in the FMO complex. We hope that the revised version is now acceptable for publication in *The Journal of Physical Chemistry Letters*.

Sincerely,

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